Adjacency spectral likelihood classification in stochastic block models

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Abstract

We study community classification from the raw degrees and a single eigensystem of a high-degree-regularized adjacency matrix. A square-root cap reweights only vertices above a large empirical degree threshold and permits arbitrarily slowly diverging expected degrees. An exact degree correction, deterministic spectral initialization and Bernoulli block likelihood updates define a finite classifier. For any fixed number of communities, comparable community sizes and a positive full-rank block profile, its expected misclassification loss is at most a constant times the true-parameter incident-row oracle risk whenever $d = np \to \infty$ and np^2 is bounded. The comparison retains both the exact Bernoulli Chernoff information and its $d^{-1/2}$ prefactor. The proof combines truncated spectral-row energy, a local scalar leave-one-out comparison and uniform fitted-score bounds. It imposes no logarithmic lower bound on expected degree.

1 Introduction

The incident edges of a vertex determine a natural likelihood classifier when the community parameters and the other labels are known. A spectral method replaces those unknown labels by a low-dimensional representation of the graph. We ask whether the degrees and a single adjacency eigensystem retain enough information to match the error of the incident-row likelihood classifier up to a constant factor.

We construct an explicit classifier with this property for stochastic block models with arbitrary fixed K. Its spectral input is a regularized adjacency matrix $\widetilde{A} = WAW$: the diagonal weight is one unless the observed degree exceeds a sufficiently large fixed multiple of the average degree. Above that threshold a square-root cap controls the row energy. The decoder retains the raw degrees, reconstructs a signed matrix with exactly those row sums, and alternates symmetric Bernoulli block fitting with ordinary likelihood classification. The only initialization is a deterministic radius-count procedure on normalized spectral rows. At least one likelihood update is required; any finite number of the specified updates is allowed.

When no degree exceeds the threshold, $W = I$, so the encoding is exactly the leading adjacency eigenpairs together with the raw degrees. For the full sparse range the theorem uses the stated cap. The weighted population signal $W\Omega W$ has informative space $\mathrm{col}(WZ)$, with one direction per true community after row normalization. This explicit adjacency construction is the version analyzed throughout the article.

The main result holds for every diverging degree scale $d = np$, subject to the upper sparsity condition $np^2 \le C_1$. In particular, it covers $d = \log\log n$ and slower divergence. The proof controls average classification risk directly, allowing isolated vertices and exceptional spectral rows. A graph-wide bound on the largest spectral row would discard this range. Instead, a truncated-energy estimate bounds the probability that the tested incident row actually encounters an exceptional coefficient.

The target comparison is sharp at the level of risk. For a pair of communities with exact Bernoulli Chernoff information I, the oracle error is of order $d^{-1/2}e^{-I}$. Sharp Chernoff bounds

in community detection are developed by Zhou and Li [4]; spectral error exponents are studied by Zhang [3]. Here the local spectral and fitted-likelihood errors are bounded on the scale needed to preserve a constant risk multiplier. The geometric input uses sparse graph regularization [2] and, at larger degree scales, a bounded-entry matrix norm inequality [1].

2 Model and the incident-row oracle

There are n vertices and a fixed number $K \geq 2$ of communities. Let $z \in [K]^n$, let Z be its indicator matrix, and let $n_a = \#\{i : z_i = a\}$. The symmetric adjacency matrix has zero diagonal and independent upper-triangular entries satisfying

$$A_{ij} \sim \mathrm{Bern}(P_{z_i z_j}), \qquad P = pC, \qquad d = np, \qquad \Omega = ZPZ^\top, \qquad i < j. \tag{1}$$

The structural constants obey

$$n_a \geq \pi_- n, \qquad C = C^\top, \qquad 0 < c_- \leq C_{ab} \leq c_+ < \infty, \qquad \sigma_{\min}(C) \geq \kappa > 0. \tag{2}$$

The number K, π_-, c_-, c_+ and κ are fixed. The matrix C may vary with n inside this class. Constants in bounds may depend on these fixed quantities and on the stated upper bound for np^2, but not on n. The completed mean Ω includes a diagonal solely for algebra; the actual mean is $\Omega - \mathrm{diag}(\Omega_{ii})$. We keep this $O(p)$ difference in the proof.

For a partition $\widehat{z}$, define

$$\ell(\widehat{z}, z) = \frac{1}{n} \min_{\sigma \in \mathfrak{S}_K} \sum_i \mathbf{1}\{\sigma(\widehat{z}_i) \neq z_i\}.$$

The oracle knows P and all labels except the tested label. At vertex i it maximizes the exact Bernoulli log likelihood

$$L_{i,*}(a) = \sum_{j \neq i} \{A_{ij} \log P_{az_j} + (1 - A_{ij}) \log(1 - P_{az_j})\}. \tag{3}$$

It uses uniform candidate weights and a fixed rule for ties. An exact global community-size constraint is not used to infer the missing label. If $z_{i,*}$ denotes this local classifier, its average directional error is

$$R_{\mathrm{SBM},*}(n) = \frac{1}{n} \sum_i \mathbb{P}\{z_{i,*} \neq z_i\}. \tag{4}$$

This is the oracle used throughout the article.

3 A decoder from degrees and adjacency spectrum

The input is A and K. Fix a sufficiently large constant $b_0 > 0$ and a sufficiently small constant $c_0 > 0$, depending only on the structural class. All choices and ties below are deterministic. A fixed fallback partition defines the algorithm whenever a required operation is undefined.

Encoding. Write

$$\mathbf{d} = A\mathbf{1}, \qquad D_i = \mathbf{d}_i, \qquad \bar{D} = \frac{1}{n} \sum_i D_i.$$

$$\tau = b_0 \bar{D}, \qquad G = \mathrm{diag}\{\max(D_i, \tau)\}, \qquad W = \tau^{1/2} G^{-1/2}, \qquad \widetilde{A} = WAW.$$

The all-zero graph has the fixed output. Otherwise G is positive, and the diagonal weights are

$$W_i = \begin{cases} 1, & D_i \leq \tau, \\ \sqrt{\tau / D_i}, & D_i > \tau. \end{cases}$$

Thus edges between vertices below the threshold keep their original weights. Retain the raw degrees and the K eigenpairs of $\widetilde{A}$ with largest absolute eigenvalues:

$$U^\top U = I_K, \qquad \widetilde{A}U = U\Xi.$$

Only $(\mathbf{d}, U, \Xi)$ are used after this step; τ, G, W are recovered from $\mathbf{d}$. All informative eigenvalues are retained by magnitude, allowing either sign.

Exact-degree reconstruction. Put

$$H = WUU^\top W^{-1}, \qquad R_0 = W^{-1}U\Xi U^\top W^{-1} = AH.$$

If the cap is inactive, these reduce to $H = UU^\top$ and $R_0 = A_K$, the rank-K adjacency spectral reconstruction. Let $v = \mathbf{d} - R_0\mathbf{1}$, let $R_1 = R_0 + v\mathbf{1}^\top/n$, and let r be the vector of diagonal entries of R_1. Define

$$R^h = R_1 + \frac{r\mathbf{1}^\top}{n-1} - \frac{n}{n-1}\operatorname{diag}(r). \tag{5}$$

Then

$$R^h\mathbf{1} = \mathbf{d}, \qquad \operatorname{diag}(R^h) = 0.$$

The matrix R^h can be signed and need not be symmetric. Its entries are not clipped. The formula is computed entirely from the retained quantities. In particular, an isolated vertex has a zero reconstructed row and does not stop classification of other vertices.

Initialization. Normalize each nonzero row of U to unit length; assign a fixed unit direction to a zero row. Among the remaining rows, select one having the largest number of remaining rows within distance $1/8$, and remove all rows within distance $1/4$ of it. Repeat for K centers, then assign every row to its nearest center. If fewer than K centers can be selected, use the fixed fallback. This is a finite procedure using $O(Kn^2)$ distance comparisons.

Block fitting and classification. Given current labels $\widehat{z}$, for $a \le b$ let $\mathcal{I}_{ab}(\widehat{z})$ contain all ordered pairs $i \ne j$ whose unordered label pair is $\{a, b\}$. Let

$$E_{ab} = \sum_{(i,j)\in\mathcal{I}_{ab}(\widehat{z})} R^h_{ij}, \qquad N_{ab} = |\mathcal{I}_{ab}(\widehat{z})|, \qquad \varepsilon = c_0\bar{D}/n.$$

The symmetric fitted matrix is

$$\widehat{P}_{ab} = \widehat{P}_{ba} = \operatorname{clip}_{[\varepsilon,\,1/2]}(E_{ab}/N_{ab}). \tag{6}$$

This maximizes the ordered-pair Bernoulli working criterion with a symmetric block probability. The fit is interior on the event used in the proof. Update all labels simultaneously by maximizing

$$\widehat{L}_i(a) = \sum_{j\ne i}\{R^h_{ij}\log\widehat{P}_{a\widehat{z}_j} + (1 - R^h_{ij})\log(1 - \widehat{P}_{a\widehat{z}_j})\}. \tag{7}$$

In particular, the full nonedge term is retained. There is no target degree parameter to profile in this ordinary SBM algorithm. Write $\widehat{L}^{ab}_i = \widehat{L}_i(a) - \widehat{L}_i(b)$ for a fitted pairwise contrast.

Perform at least one block fit and likelihood update. The output may follow any finite number of these hard refits, including finite data-dependent stopping. Empty fitted groups or undefined operations use the stated fallback; the proof shows they do not occur on the aggregate event. No leave-one-out graph, true label or true model parameter is an algorithm input.

4 Oracle-risk guarantee

Theorem 4.1. *Under (1)–(2), suppose*

$$d = np \longrightarrow \infty, \qquad np^2 \le C_1 < \infty.$$

For sufficiently large fixed b_0 and sufficiently small fixed $c_0 > 0$, the algorithm in section 3, after any finite positive number of the specified likelihood updates, satisfies

$$\mathbb{E}\ell(\widehat{z}, z) \le C R_{\mathrm{SBM},*}(n) \tag{8}$$

for all sufficiently large n. The same bound holds for finite data-dependent stopping. The constant C is independent of n and of the number of refits.

There is no condition of the form $np \ge c \log n$. The assumptions allow general symmetric positive block profiles and comparable, unequal community sizes. They do not require a planted partition or equal expected degrees. The condition $np^2 \le C_1$ is retained as an upper sparsity assumption.

For the exact pairwise Bernoulli information $I_{i,ab}$ defined in (14), lemma A.1 gives

$$R_{\mathrm{SBM},*}(n) \asymp \frac{1}{n\sqrt{d}} \sum_i \sum_{b \ne z_i} e^{-I_{i,z_i b}}.$$

Thus (8) preserves the sharp prefactor as well as the exponent. It asserts a bounded risk ratio; it does not assert that the ratio tends to one. Since a nonzero loss is at least $1/n$, the theorem also implies exact recovery whenever $n R_{\mathrm{SBM},*}(n) \to 0$.

4.1 Proof structure

The proof has four ingredients. First, regularized adjacency geometry gives an informative spectral gap of order d, $O(d^{-1/2})$ subspace error and $O(n)$ weighted cap energy $\sum_i (G_i - \tau)^2 / G_i$. The threshold lies above all population degrees on the aggregate event, so this energy is bounded by the weighted degree fluctuation. These aggregate bounds suffice for the deterministic initialization and an invariant $O(n/d)$ misclassification basin.

Second, a same-threshold truncated-energy recursion controls large spectral rows in expectation. By taking a fixed row threshold large enough, the average probability of an exceptional tested row, or of an incident edge hitting an exceptional training row, is at most $C_H e^{-Hd}$ for any prescribed fixed H. This estimate replaces a simultaneous row bound.

Third, on the local regular event, the exact block fit has error $O(1/n)$, and every fitted row contrast differs by $O(1)$ from its true block contrast. A scalar leave-one-out identity replaces the full-graph spectral action by a function of independent training data and the tested incident row, again with $O(1)$ error. Under the oracle Chernoff tilt, the resulting coefficient error has bounded mean and quadratic mass.

Finally, a sharp Bernoulli transfer preserves $d^{-1/2} e^{-I_{i,ab}}$. Exceptional coefficients are paid for only when the incident row hits them. Polynomially accurate regularized geometry suffices below an internal logarithmic proof threshold; an exponentially accurate norm event supplies the complementary range. These estimates together cover every $d \to \infty$. The appendices give the complete argument and the proof of theorem 4.1.

Organization of related material. The companion notes contain the degree-normalized formulation, degree correction, alternative earlier decoders, counterexamples and supplementary derivations. None of those notes is required for this article's theorem or proof.

A A sharp Bernoulli likelihood transfer

We give the probability step explicitly to distinguish a risk factor from an exponential-rate factor. Sharp Chernoff comparisons in community detection are also developed by Zhou and Li [4]. The argument below allows a perturbation measurable from independent training data.

Lemma A.1 (Bounded tilted error preserves the sharp prefactor)**.** *Let X_j be independent Bernoulli variables with probabilities p_j, and let*

$$T = b_0 + \sum_j w_j X_j, \qquad K(s) = \log \mathbb{E} e^{-sT}.$$

Suppose $|w_j| \le W$, the minimizer s_* satisfies $\tau \le s_* \le 1 - \tau$ for a fixed $\tau > 0$, and under the probability measure Q proportional to $e^{-s_* T}$,

$$\mathbb{E}_Q T = 0, \quad vd \le \mathrm{Var}_Q(T) \le V_0 d, \quad \sum_j r_j \le V_0 d, \tag{9}$$

where $r_j = Q(X_j = 1)$. Put $I = -K(s_*)$. For every fixed $L < \infty$ and all sufficiently large d, uniformly over $|x| \le L$,

$$c_L d^{-1/2} e^{-I} \le \mathbb{P}(T < x) \le \mathbb{P}(T \le x) \le C_L d^{-1/2} e^{-I}. \tag{10}$$

Let $\mathcal{W}$ be independent of $(X_j)_j$ and let

$$\widehat{T} = T + b + \sum_j e_j X_j$$

have coefficients measurable with respect to $\mathcal{W}$. On a measurable training event suppose

$$\max_j |e_j| \le M, \qquad \left| b + \sum_j r_j e_j \right| \le B, \qquad \sum_j r_j e_j^2 \le V. \tag{11}$$

For fixed M, V and every real x,

$$\mathbb{P}(\widehat{T} \le x \mid \mathcal{W}) \le C \exp\{s_* x + B + e^M V/2\}\, d^{-1/2} e^{-I}. \tag{12}$$

The constant is uniform over the training event. In particular B can be a nonnegative training-measurable random bound, provided its indicated exponential moment is integrated when removing the conditioning.

Proof. Under Q, the coordinates remain independent Bernoulli variables,

$$r_j = \frac{p_j e^{-s_* w_j}}{1 - p_j + p_j e^{-s_* w_j}}.$$

The sum of their centered weighted third absolute moments is at most $2W^3 \sum_j r_j = O(d)$. Berry–Esseen and (9) therefore show that every interval of length one has Q probability at most $C/\sqrt{d}$, uniformly in its location. Since $d\mathbb{P}/dQ = e^{-I} e^{s_* T}$, partitioning $(-\infty, x]$ into unit intervals gives the upper bound in (10).

For the lower bound choose a fixed interval $(x - L_0, x)$, with L_0 sufficiently large. Its Gaussian probability under a centered normal variable of variance in $[vd, V_0 d]$ is at least $cL_0/\sqrt{d}$ for large d, uniformly over bounded x. Choose L_0 so this exceeds twice the Berry–Esseen error. The Q probability is then at least $c'/\sqrt{d}$, and $e^{s_* T}$ is bounded below by a positive constant on the interval. Using strict endpoints also handles arbitrary lattice atoms and tie rules.

For the perturbation, independence of training and the row allows us to regard b, e as fixed. The elementary Bernoulli exponential-moment bound gives

$$\begin{aligned}\log \mathbb{E} e^{-s_* \widehat{T}} &= -I - s_* b + \log \mathbb{E}_Q e^{-s_* \sum_j e_j X_j} \\ &\le -I - s_* \left(b + \sum_j r_j e_j \right) + \frac{s_*^2 e^{s_* M}}{2} \sum_j r_j e_j^2 \\ &\le -I + B + e^M V/2.\end{aligned}$$

Now tilt the original law by $e^{-s_* \widehat{T}}$. Under this new measure $\widehat{Q}$ the coordinates remain independent and their odds are those under Q multiplied by $e^{-s_* e_j}$. Consequently their probabilities and variances are within fixed factors of those under Q. Furthermore,

$$\left(\sum_j r_j (1 - r_j)(w_j + e_j)^2 \right)^{1/2} \ge \sqrt{vd} - \sqrt{V}.$$

For large d, the variance of $\widehat{T}$ under $\widehat{Q}$ is therefore between positive constant multiples of d. Its centered third absolute moments sum to $O(d)$ because $w_j + e_j$ are uniformly bounded and $\sum_j \widehat{r}_j = O(d)$. Berry–Esseen bounds every unit interval under $\widehat{Q}$ by $C/\sqrt{d}$, irrespective of its mean. The same interval summation, with the displayed exponential moment, proves (12). □

A.1 Application to the actual Bernoulli SBM oracle

For a true label a and a competitor b, the oracle contrast is

$$T_{i,ab} = L_{i,*}(a) - L_{i,*}(b) = b_{i,ab} + \sum_{j \ne i} w_{z_j}^{ab} A_{ij},$$

where

$$w_c^{ab} = \log \frac{P_{ac}(1 - P_{bc})}{P_{bc}(1 - P_{ac})}, \qquad b_{i,ab} = \sum_{j \ne i} \log \frac{1 - P_{az_j}}{1 - P_{bz_j}}. \tag{13}$$

Its exact Chernoff information is

$$I_{i,ab} = - \min_{0 \le s \le 1} \sum_{j \ne i} \log \left[P_{az_j}^{1-s} P_{bz_j}^{s} + (1 - P_{az_j})^{1-s} (1 - P_{bz_j})^{s} \right]. \tag{14}$$

On the compact positive full-rank classes used here, probabilities are comparable to p and bounded away from one, the weights are bounded, s_* stays away from the endpoints, and the tilted variance is comparable to d. To verify the last assertions, the derivatives of the transform at its endpoints are respectively minus and plus the two directional Bernoulli KL divergences, both of order d. Its second derivative is bounded by Cd. The row separation implied by $\sigma_{\min}(P) \ge \kappa p$ gives a lower bound cd on that derivative throughout $[0, 1]$: tilted probabilities are comparable to p and the sum of squared log-odds differences is at least cn. Thus lemma A.1 applies uniformly to every pair.

A.2 Fixed-K comparison and label alignment

Let q_i be the error probability of the specified oracle at vertex i. For any coherent family of its candidate scores, strict pairwise loss to a competitor implies an oracle error; an oracle error implies non-strict loss to some competitor. Hence

$$\max_{b \ne a} \mathbb{P}(T_{i,ab} < 0) \le q_i \le \sum_{b \ne a} \mathbb{P}(T_{i,ab} \le 0).$$

By (10),

$$q_i \asymp \frac{1}{\sqrt{d}} \sum_{b \neq z_i} e^{-I_{i,ab}} \tag{15}$$

The constants may include a factor depending on fixed K. Fitted partitions are aligned with the true groups by a single common permutation on the good event. Uniform score comparisons under that alignment imply

$$\mathbb{E}[\ell(\hat{z}, z)\mathbf{1}\{\mathcal{G}\}] \leq \frac{1}{n} \sum_i \mathbb{P}(\text{aligned fitted error at } i, \mathcal{G}).$$

The ensuing pairwise bounds apply to fixed true coefficient vectors; a data-dependent matching does not become an independence assumption.

A.3 Exceptional coefficients

Lemma A.2 (Paying exceptional coefficients only when the row hits them)**.** *Assume the oracle hypotheses of lemma A.1, and write p_j for the un-tilted row probabilities. Suppose $p_j \leq C_p r_j$, as holds here for bounded oracle likelihood coefficients. On an independent training event $\mathcal{E}$, suppose*

$$\left| b + \sum_j r_j e_j \right| \leq B, \qquad \sum_j r_j e_j^2 \leq V.$$

Here B, V are fixed constants. For any fixed $M > 0$ define

$$h_M = \sum_{j:|e_j|>M} p_j, \qquad \eta_M = \sum_j r_j e_j^2 \mathbf{1}\{|e_j| > M\}.$$

Then, for every fixed bounded x,

$$\mathbb{P}(\widehat{T} \leq x) \leq C_M d^{-1/2} e^{-I} + \mathbb{E}[\mathbf{1}\{\mathcal{E}\} h_M] + \mathbb{P}(\mathcal{E}^c). \tag{16}$$

Since $h_M \leq C_p \eta_M / M^2$, an expected truncated quadratic error of order $d^{-1/2} e^{-I}$ suffices. A bound Ce^{-Hd} is a convenient sufficient condition whenever $I \leq C_I d$ and $H > C_I$. The expected hit-mass condition on h_M is weaker and is itself enough.

Proof. For analysis only, replace e_j by $e'_j = e_j \mathbf{1}\{|e_j| \leq M\}$. The truncated coefficients satisfy

$$\left| b + \sum_j r_j e'_j \right| \leq B + \eta_M / M \leq B + V/M, \qquad \sum_j r_j (e'_j)^2 \leq V.$$

Hence lemma A.1 bounds their score error by $C_M d^{-1/2} e^{-I}$ on $\mathcal{E}$. The original and truncated random scores are identical whenever $X_j = 0$ for every j with $|e_j| > M$. Conditional on training, the probability that this fails is at most

$$\sum_{j:|e_j|>M} p_j = h_M \leq \frac{C_p}{M^2} \eta_M.$$

Integrate this bound on $\mathcal{E}$ and pay at most $\mathbb{P}(\mathcal{E}^c)$ outside it. No coefficient truncation is an additional input or computation required of the implemented decoder. □

B Aggregate geometry for the capped adjacency matrix

Let $D_0 = \mathrm{diag}(\Omega\mathbf{1})$, using the completed mean in (1). Its diagonal entries are block constant and lie between fixed positive multiples of d. Throughout this section $\tau = b_0\bar{D}$, $G_i = \max(D_i, \tau)$ and $W = \tau^{1/2}G^{-1/2}$ are the high-degree cap used by the algorithm, with b_0 sufficiently large for the structural class. Thus $\widetilde{A} = WAW$ leaves every edge between vertices of degree at most τ unchanged. For analysis only, write $Q = \widetilde{A}/\tau = G^{-1/2}AG^{-1/2}$ and $\|v\|_G^2 = v^\top Gv$. The scalar rescaling Q has the same eigenvectors as the encoded adjacency matrix $\widetilde{A}$.

Proposition B.1 (Aggregate capped-adjacency geometry)**.** *Under (2), let $d = np \to \infty$ and $np^2 \le C_1$. For every fixed $r > 0$, with probability at least $1 - C_r n^{-r}$, $cd \le \tau \le Cd$, $\tau \ge \max_i (D_0)_{ii}$, $cnd \le \sum_i D_i$, and*

$$\|W(A-\Omega)W\|_{\mathrm{op}} \le C_r\sqrt{d}, \qquad \sum_i G_i \le C_r nd, \tag{17}$$

$$\sum_i \frac{(D_i - (D_0)_{ii})^2}{G_i} + \sum_i \frac{(G_i - \tau)^2}{G_i} \le C_r n. \tag{18}$$

In particular, $\left\|G^{-1/2}(A-\Omega)G^{-1/2}\right\|_{\mathrm{op}} \le C_r/\sqrt{d}$. Let U contain the K eigenvectors of $\widetilde{A}$ with largest absolute eigenvalues, and put $\Pi = UU^\top$ and $H = G^{-1/2}\Pi G^{1/2}$. Its retained eigenvalues have magnitude at least $c_r d$; the remaining eigenvalues have magnitude at most $C_r\sqrt{d}$. There is an orthogonal matrix O such that the orthonormal population basis

$$V = G^{-1/2}Z(Z^\top G^{-1}Z)^{-1/2}O$$

satisfies

$$\|U - V\|_F \le C_r/\sqrt{d}, \qquad \sqrt{G_i}\,\|V_{i\cdot}\|_2 \le C_r\sqrt{d/n}.$$

Moreover,

$$\left\|(I-\Pi)G^{1/2}Z\right\|_F \le C_r\sqrt{n}, \tag{19}$$

and every block-constant f with $\|f\|_\infty \le C_f$ obeys

$$\|(I-H)f\|_G \le C_r C_f\sqrt{n}, \qquad \|(I-H)f\|_2^2 \le C_r C_f^2 n/d. \tag{20}$$

The same assertions hold simultaneously after deletion of any fixed number of vertices, with each cap recomputed from the remaining graph, after increasing the probability parameter. No raw minimum-degree condition is imposed.

Proof. The Bernoulli transform of the total number of edges gives $cnd \le \sum_i D_i \le Cnd$ except with probability e^{-cnd}. For large enough b_0, this event implies

$$cd \le \tau = b_0\bar{D} \le Cd, \qquad \tau \ge \max_i (D_0)_{ii}.$$

Let $d_* = \max_{i\ne j} n\mathbb{E}A_{ij} \asymp d$, and put $W_{h,ii} = \sqrt{\min(1, 2d_*/D_i)}$, with value one at zero degrees. The changed set $\mathcal{S} = \{i : D_i > 2d_*\}$ contains at most $10n/d_*$ vertices except with probability e^{-cn}. To check the strength of this bound, set $m = \lfloor 10n/d_* \rfloor + 1$. If $m \le n$ and m vertices exceed the threshold, some set T of size m has $\sum_{i\in T} D_i > 2md_*$. This is an independent-edge sum with coefficients in $\{1, 2\}$ and mean at most md_*. At $t = (\log 2)/2$, its exponential transform bounds the tail by

$$\exp\{-(\log 2 - 1/2)md_*\},$$

because $(e^{tc} - 1)/c \le (e^{2t} - 1)/2 = 1/2$ for $c = 1, 2$. Since $\log\binom{n}{m} \le m\log(en/m) = o(md_*)$, a union over T gives the stated bound. If $m > n$, the claim is automatic.

Every row of W_hAW_h has squared Euclidean norm at most $W_{h,ii}^2 D_i \le 2d_*$. The row-norm form of the regularized adjacency theorem of Le, Levina and Vershynin [2, Theorem 2.1 and Remark 2.2] therefore yields

$$\|W_hAW_h - \mathbb{E}A\|_{\mathrm{op}} \le C_r\sqrt{d}$$

with failure probability at most $C_r n^{-r}$. The theorem permits these graph-dependent weights. Also

$$\|W_h\Omega W_h - \Omega\|_{\mathrm{op}} \le Cp\sqrt{n|\mathcal{S}|} \le C\sqrt{d},$$

by the Frobenius norm: the difference is supported on rows or columns in $\mathcal{S}$. The completed-mean diagonal has norm $O(p)$. Consequently $\|W_h(A-\Omega)W_h\|_{\mathrm{op}} \le C_r\sqrt{d}$. There is a bounded diagonal F such that

$$G^{-1/2} = (2d_*)^{-1/2}FW_h, \qquad F_{ii} = \sqrt{\frac{\max(D_i, 2d_*)}{G_i}}.$$

Indeed, $G_i \ge \max(D_i, cd)$ and $d_* \asymp d$. Conjugation gives the stated bound for the noise in Q; multiplication by $\tau \asymp d$ proves the adjacency-scale bound in (17). Since $G_i \le D_i + \tau$, the total weight $\sum_i G_i$ is at most Cnd.

Applying the scalar-rescaled noise to $G^{1/2}\mathbf{1}$ gives

$$\left\|G^{-1/2}(A-\Omega)\mathbf{1}\right\|_2 \le C_r d^{-1/2}\left\|G^{1/2}\mathbf{1}\right\|_2 \le C_r\sqrt{n},$$

which is the first sum in (18). For the second sum, $G_i - \tau = 0$ when $D_i \le \tau$; otherwise $0 \le G_i - \tau = D_i - \tau \le D_i - (D_0)_{ii}$. Thus, coordinatewise,

$$\frac{(G_i-\tau)^2}{G_i} \le \frac{(D_i-(D_0)_{ii})^2}{G_i}.$$

The same bound also gives $\sum_i (1-W_{ii})^2 \le C_r n/d$, since $(1-W_{ii})^2 \le (G_i-\tau)^2/(\tau G_i)$.

Let $J = Z^\top G^{-1}Z$. For every block a, Cauchy–Schwarz, comparable group sizes, $G_i \ge cd$ and the total-degree bound give

$$\frac{cn}{d} \le \frac{n_a^2}{\sum_{i:z_i=a} G_i} \le J_{aa} = \sum_{i:z_i=a} G_i^{-1} \le \frac{Cn}{d}.$$

The nonzero spectrum of the capped population matrix $W\Omega W$ is the spectrum of $p\tau J^{1/2}CJ^{1/2}$. Full rank of C gives a separation of order d from zero. The adjacency noise bound and subspace perturbation then give the stated eigenspace and eigenvalue bounds. The formula for V and the lower bound on J give its weighted row bound.

The matrix $\tau G^{-1/2}Z$ belongs to the population signal space $\mathrm{col}(WZ) = \mathrm{col}(G^{-1/2}Z)$. Furthermore,

$$\left\|G^{1/2}Z - \tau G^{-1/2}Z\right\|_F^2 = \sum_i \frac{(G_i-\tau)^2}{G_i} \le C_r n, \qquad \left\|G^{1/2}Z\right\|_F^2 \le C_r nd.$$

Together with the projector error $O(d^{-1/2})$, these imply (19). The matrix H is the G-orthogonal projector corresponding to Π, so the first bound in (20) follows by writing $f = Zb$. The second uses $G_i \ge cd$.

Deleting a fixed number of vertices preserves the structural bounds. Apply the same argument to each recomputed graph and choose a larger fixed probability parameter before the finite-depth union bound. □

C Truncated spectral-row energy

The next lemma controls large spectral rows in expectation. Its proof uses a recurrence at a single truncation threshold, together with concentration of independent deleted-row innovations.

Work under the fixed structural assumptions of proposition B.1, with $d = np \to \infty$ and $np^2 \le C_1$. For the high-degree cap in section 3, recall

$$\tau = b_0 \bar{D}, \qquad G_j = \max(D_j, \tau), \qquad W = \tau^{1/2} G^{-1/2}, \qquad \widetilde{A} = WAW.$$

Let U be an orthonormal basis of the informative K-dimensional eigenspace and write $\widetilde{A}U = U\Xi$. All bounds below are uniform in the fixed compact structural class. In particular,

$$0 \le \Omega_{ij} \le Cd/n, \qquad \|\Omega_{i\cdot}\|_2 \le Cd/\sqrt{n}. \tag{21}$$

Fix a probability parameter $r > 0$. Apply proposition B.1 with a sufficiently larger fixed parameter to the full graph and all graphs obtained by deleting at most two vertices. Let $\mathcal{E}_r$ be the intersection of the resulting geometry and two-sided total-degree events. Restricting the population matrix preserves the fixed structural bounds and comparable block masses. Thus

$$\mathbb{P}(\mathcal{E}_r^c) \le C_r n^{-r}. \tag{22}$$

On this event, for every graph in the deletion family, its recomputed threshold is comparable to d, its total degree is at most $C_r nd$, its centered weighted noise has norm at most $C_r\sqrt{d}$, its informative eigenvalues have absolute values at least $c_r d$, and its remaining eigenvalues have absolute values at most $C_r\sqrt{d}$. For a deleted training graph, use the analogous event formed from that graph and its further deletions. This event is measurable from the training graph alone and has the same probability bound.

The following deterministic properties of the cap will be useful:

$$W_j \le 1, \qquad W_j^2 D_j \le \tau, \qquad \left\|\widetilde{A}\right\|_{\mathrm{op}} \le \tau. \tag{23}$$

Indeed, $G \ge \mathrm{diag}(A\mathbf{1})$ and the inequality $2|ab| \le a^2 + b^2$ gives $|x^\top Q x| \le \sum_j (D_j/G_j) x_j^2 \le \|x\|_2^2$, where $Q = \widetilde{A}/\tau$.

Lemma C.1 (Truncated energy for the capped-adjacency eigenvectors)**.** *There are fixed constants $R_0, C, c > 0$ such that, for every fixed $R \ge R_0$ and all sufficiently large n,*

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}_r\} \sum_{j=1}^{n} \|U_{j\cdot}\|_2^2 \, \mathbf{1}\{\sqrt{n}\,\|U_{j\cdot}\|_2 > R\}\right] \le CR^2 e^{-cd\log R}. \tag{24}$$

The constants R_0, C, c depend on the fixed structural bounds, b_0, and r, but not on R or n. The sufficiently-large-n condition may depend on the chosen fixed R. The same result holds for each recomputed deleted graph with its training-measurable geometry event.

Proof. We give the weight perturbation, the same-threshold energy recurrence, and its expectation bounds separately.

Recomputed weights and the eigenspace residual. For $J = [n] \setminus \{i\}$, let $W^{(-i)}$ and $U^{(-i)}$ be computed from A_{JJ}, with $U^{(-i)}$ embedded by a zero row at i. Outside the geometry event choose an arbitrary deterministic orthonormal fallback depending only on A_{JJ}. For an all-zero training graph set $W^{(-i)} = I$ and use that orthonormal fallback. Thus the proof-only completion always has $\left\|W^{(-i)}\right\|_{\mathrm{op}} \le 1$, and the independent innovation below is defined for every training graph. Put $(\Gamma_i)_{jj} = W_j^{(-i)}/W_j - 1$ for $j \in J$ and $(\Gamma_i)_{ii} = 0$. The exact threshold difference is

$$\tau^{(-i)} - \tau = b_0 \frac{\bar{D} - 2D_i}{n-1}.$$

Comparing the two square-root caps, using the comparability of the thresholds to d, gives on $\mathcal{E}_r$

$$|(\Gamma_i)_{jj}| \le C\left\{\frac{A_{ij}}{G_j} + \frac{d+D_i}{nd}\right\}, \qquad \|\Gamma_i\|_{\mathrm{op}} \le C/d. \tag{25}$$

This also covers a crossing of the cap threshold. It follows either by splitting the cases in the two maxima or by comparing the logarithms of $\sqrt{\tau/\max(D_j,\tau)}$; the relative threshold change is $O((d+D_i)/(nd))$. The last bound uses $D_i \le n$ and $d \le Cn$.

There is an orthogonal matrix O_i such that

$$\left\|U - U^{(-i)}O_i\right\|_F \le C\,\|U_{i\cdot}\|_2 + C\,\|\Gamma_i U\|_F\,. \tag{26}$$

To verify this, write $\widetilde{A} = \left(\begin{smallmatrix} 0 & m_i^\top \\ m_i & B_i \end{smallmatrix}\right)$. By (23), $\|m_i\|_2 \le \sqrt{\tau}$ and $\|B_i\|_{\mathrm{op}} \le \tau$. Also $B_iU_J = U_J\Xi - m_iU_{i\cdot}$. The lower block of the recomputed deleted matrix is $(I+\Gamma_i)B_i(I+\Gamma_i)$. Subtracting the full matrix and applying the difference to U gives a Frobenius residual bounded by $Cd(\|U_{i\cdot}\|_2 + \|\Gamma_iU\|_F)$. The full informative spectrum is separated from the deleted noninformative spectrum by cd. The complementary-space Sylvester equation and orthogonal Procrustes alignment give (26). Only separation from the complementary spectrum is used; the informative eigenvalues can repeat or have either sign.

Truncation and summable weight changes. Set $t = R/\sqrt{n}$. Let f_t be radial projection onto the Euclidean ball of radius t, applied to each row, and put

$$V = f_t(U), \qquad T = U - V, \qquad H_R(U) = \sum_j \|U_{j\cdot}\|_2^2\,\mathbf{1}\{\|U_{j\cdot}\|_2 > t\}.$$

The radial projection is 1-Lipschitz and commutes with orthogonal rotations. Moreover $\|T\|_F^2 \le H_R(U)$, $\max_j \|V_{j\cdot}\|_2 \le t$, and $\|V\|_F \le \sqrt{K}$. Define

$$\beta_i = W_i\left(\sqrt{D_i} + Cd/\sqrt{n}\right).$$

By (23), $\beta_i^2 \le Cd$. Equation (25) yields

$$\|\Gamma_iV\|_F \le Ct\frac{\sqrt{D_i}}{d} + C\frac{d+D_i}{nd},$$

and therefore

$$\frac{C\beta_i}{d}\,\|\Gamma_iV\|_F \le Ct\left\{\frac{1}{d} + \frac{\sqrt{D_i}}{d^{3/2}}\right\}. \tag{27}$$

For the global term, use $t \ge n^{-1/2}$, $D_i \le n$, and $d \le Cn$: $D_i/(nd^{3/2}) \le t\sqrt{D_i}/d^{3/2}$ and $1/(n\sqrt{d}) \le Ct/d$. The square-root cap does not give $W_iD_i \le Cd$. The right side of (27) can exceed a sufficiently small fixed multiple of t, for large d, only if $D_i \ge c_*d^3$.

The unbounded part satisfies the summable estimate

$$\sum_i \beta_i^2\,\|\Gamma_iT\|_F^2 \le C\,\|T\|_F^2\,. \tag{28}$$

Indeed, after interchanging i,j, the local coefficient of $\|T_{j\cdot}\|_2^2$ is at most $CdD_j/G_j^2 \le C$. The global coefficient is bounded by

$$\frac{C}{n^2d}\sum_i (d+D_i)^2 \le C,$$

because $\sum_i D_i \le Cnd$ and $\sum_i D_i^2 \le n\sum_i D_i$. These estimates do not use a maximum-degree event.

Independent innovations and the recurrence. Define the unrotated, training-measurable-coefficient innovation

$$K_i = (A_{iJ} - \Omega_{iJ})W^{(-i)} f_t(U_J^{(-i)}). \tag{29}$$

Let $E_0 = W(A - \Omega)W$. The eigenvector equation is

$$U = W\Omega WU\Xi^{-1} + E_0 V\Xi^{-1} + E_0 T\Xi^{-1}.$$

The first term has every row bounded by $C/\sqrt{n}$, by (21). For the second, compare $W_J V_J$ to $W^{(-i)} f_t(U_J^{(-i)})O_i$ and use radial Lipschitzness, (26), and (25). This gives

$$\begin{aligned}\|U_{i\cdot}\|_2 \le C/\sqrt{n} + (C/d)\,\|K_i\|_2 + (C\beta_i/d)\,\|U_{i\cdot}\|_2 \\ + (C\beta_i/d)\,\|\Gamma_i V\|_F + (C\beta_i/d)\,\|\Gamma_i T\|_F + (C/d)\,\|(E_0T)_{i\cdot}\|_2\,.\end{aligned} \tag{30}$$

The completed-mean diagonal contributes at most $Cpt/d \le Ct/n$ and is absorbed in the bounded terms; it is not treated as an observed self-loop. Although O_i may depend on the tested row, the argument only uses $\|K_iO_i\|_2 = \|K_i\|_2$. No conditional concentration is applied to rotated coordinates.

Take R sufficiently large and then d sufficiently large to absorb the baseline and self terms. Retain from (27) the excess $Ct\sqrt{D_i/d^3}\mathbf{1}\{D_i \ge c_*d^3\}$. For $\|U_{i\cdot}\|_2 > t$, separate the cases $\|K_i\|_2 > c'dt$ and $\|K_i\|_2 \le c'dt$ and square. The resulting bound is

$$\begin{aligned}\|U_{i\cdot}\|_2^2\,\mathbf{1}\{\|U_{i\cdot}\|_2 > t\} \le \frac{C}{d^2}\,\|K_i\|_2^2\,\mathbf{1}\{\|K_i\|_2 > c'dt\} \\ + \frac{C}{d^2}\,\|(E_0T)_{i\cdot}\|_2^2 + \frac{C\beta_i^2}{d^2}\,\|\Gamma_i T\|_F^2 + \frac{Ct^2}{d^3}D_i\mathbf{1}\{D_i \ge c_*d^3\}.\end{aligned} \tag{31}$$

Summing, using $\|E_0\|_{\mathrm{op}}^2 \le Cd$ and (28), and multiplying by $\mathbf{1}\{\mathcal{E}_r\}$ gives

$$\begin{aligned}H_R(U)\mathbf{1}\{\mathcal{E}_r\} \le \frac{C}{d^2}\sum_i \|K_i\|_2^2\,\mathbf{1}\{\|K_i\|_2 > c'dt\} + \frac{C}{d}H_R(U)\mathbf{1}\{\mathcal{E}_r\} \\ + \frac{Ct^2}{d^3}\sum_i D_i\mathbf{1}\{D_i \ge c_*d^3\}.\end{aligned} \tag{32}$$

Both energy terms use the same threshold R.

Expectation bounds. Conditional on A_{JJ}, (29) is a sum of independent centered vectors. In each coordinate its summands are bounded by t, and its total variance is at most

$$Cp\left\|W^{(-i)} f_t(U^{(-i)})\right\|_F^2 \le Cd/n.$$

These bounds hold for every training graph, without conditioning on $\mathcal{E}_r$. Bennett's inequality in the fixed K coordinates, followed by integration of the squared-norm tail, gives

$$\mathbb{E}\Big[\|K_i\|_2^2\,\mathbf{1}\{\|K_i\|_2 > c'dt\}\Big] \le Cd^2t^2e^{-c_1 d\log R}. \tag{33}$$

To see the exponent, set $v = Cd/n$ and $h(u) = (1+u)\log(1+u) - u$. At $x = c'dt$, the Bennett exponent $(v/t^2)h(tx/v)$ is at least $c_1 d\log R$, since tx/v is proportional to R^2. Beyond that point the derivative of the exponent is at least $c\log R/t$, which gives the same bound upon tail integration.

Every D_i is a Bernoulli sum with mean at most Cd. Its Chernoff bound and tail integration yield

$$\mathbb{E}[D_i\mathbf{1}\{D_i \ge c_*d^3\}] \le Cd^3e^{-c_2d^3\log d}. \tag{34}$$

The left side is zero if $c_* d^3 > n$. Taking expectations in (32), absorbing C/d, and using $nt^2 = R^2$ now gives

$$\mathbb{E}[H_R(U)\mathbf{1}\{\mathcal{E}_r\}] \le CR^2\{e^{-c_1 d \log R} + e^{-c_2 d^3 \log d}\}.$$

For each fixed R, the second exponential is eventually bounded by the first after reducing c_1. This proves (24). The proof is uniform under deletion of a fixed number of vertices and uses their recomputed weights, so it also proves the asserted training-graph version. The same proof applies to any event supplying the stated aggregate geometry, including the exponentially accurate event of lemma F.1. The innovation and degree-tail bounds are unconditional before restriction to that event. □

Corollary C.2 (Exceptional coefficients are hit with exponentially small mass)**.** *On a training graph with the preceding cap, put $H = G^{-1/2}UU^\top G^{1/2}$. Let f be any training-measurable vector satisfying $\|f\|_\infty \le F_0$, and let $e = (H - I)f$. For every prescribed fixed $H_0 > 0$, there is a fixed $M < \infty$ such that, for any independent tested Bernoulli row with probabilities $p_j \le Cd/n$,*

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}_r\} \sum_{j:|e_j|>M} p_j\right] \le C_{H_0} e^{-H_0 d}. \tag{35}$$

Here $\mathcal{E}_r$ is the training-measurable geometry event.

Proof. On that event, $G_j \ge a_- d$ and $\sum_j G_j \le C_G n d$. Consequently

$$|e_j| \le F_0 + \frac{\|U_{j\cdot}\|_2}{\sqrt{G_j}} \left\|U^\top G^{1/2} f\right\|_2 \le F_0 + F_0\sqrt{C_G/a_-}\sqrt{n}\,\|U_{j\cdot}\|_2 .$$

For $M_R = F_0 + F_0\sqrt{C_G/a_-}R$, the event $|e_j| > M_R$ implies $\sqrt{n}\,\|U_{j\cdot}\|_2 > R$. Thus lemma C.1 gives

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}_r\}\sum_j \mathbf{1}\{|e_j| > M_R\}\right] \le \frac{n}{R^2}\mathbb{E}[H_R(U)\mathbf{1}\{\mathcal{E}_r\}] \le Cne^{-cd\log R}. \tag{36}$$

Multiplication by Cd/n bounds the expected hit mass by $Cde^{-cd\log R}$. Choose a fixed R with $c\log R > H_0 + 1$ and absorb the factor d. Independence is required only to interpret the conditional probability that the tested row hits the training exceptional set; no independence between the training geometry event and its eigenvectors is asserted. □

Corollary C.3 (Average exceptional target-row mass)**.** *For every fixed degree cutoff L and every fixed $H_0 > 0$, there is a fixed L_0 such that*

$$\frac{1}{n}\sum_i \mathbb{P}\left\{\sqrt{G_i}\,\|U_{i\cdot}\|_2 > L_0\sqrt{d/n},\quad D_i \le Ld,\quad \mathcal{E}_r\right\} \le C_{H_0,L}e^{-H_0 d}. \tag{37}$$

Proof. On the displayed event, $G_i \le C_L d$. Choose $L_0 = R\sqrt{C_L}$. The target-row exception then implies $\sqrt{n}\,\|U_{i\cdot}\|_2 > R$. Apply (24) and divide its corresponding expected count by n. Choosing fixed R sufficiently large gives (37). □

D Local stability under deletion of the tested vertex

The comparison below requires spectral row bounds only on neighbors of the tested vertex. The cap threshold is recomputed after deletion. We divide each capped adjacency matrix by its own scalar threshold to write a generalized eigenvalue equation; this does not change its retained eigenspace or the reconstruction AH.

Lemma D.1 (A local scalar comparison for capped adjacency)**.** *Write* $A = \left(\begin{smallmatrix} 0 & x^\top \\ x & B \end{smallmatrix}\right)$, $k = \sum_j x_j \le Ld$, *and let* $\Delta = \mathrm{diag}\{\max(D_{B,j}, \tau_B)\}$ *be recomputed from* B. *Suppose*

$$cd \le \Delta_j, \qquad \sum_j \Delta_j \le Cnd, \qquad Q_B = \Delta^{-1/2} B \Delta^{-1/2} = U_B \Lambda_B U_B^\top + N,$$

where $NU_B = 0$, $\|N\|_{\mathrm{op}} \le C/\sqrt{d}$, *and the* K *retained eigenvalues have a fixed gap from zero. Let the full diagonal cap weight be* $G = \mathrm{diag}(g, \Delta + E)$, *with*

$$cd \le g \le Cd, \qquad E = E_{\mathrm{loc}} + E_{\mathrm{glob}}, \qquad 0 \le E_{\mathrm{loc}} \le \mathrm{diag}(x), \quad \|E_{\mathrm{glob}}\|_{\mathrm{op}} \le Cd/n.$$

Here $d \to \infty$ *and* $d \le Cn$. *Suppose only on hit coordinates that*

$$\left\| \Delta_j^{-1/2} (U_B)_{j\cdot} \right\|_2 \le R/\sqrt{nd} \quad \textit{whenever } x_j = 1. \tag{38}$$

Let U *contain orthonormal informative eigenvectors of* $\widetilde{A} = WAW$, *and put*

$$H_G = G^{-1/2} U U^\top G^{1/2}, \qquad H_B = \Delta^{-1/2} U_B U_B^\top \Delta^{1/2}.$$

For every $\|w\|_\infty \le W_0$, *uniformly over the stated inputs,*

$$(AH_G w)_i = x^\top H_B w_{-i} + O_{L,R,W_0}(1). \tag{39}$$

No bound is assumed on unhit rows of U_B *or on* $\max_j \Delta_j$.

Proof. For $k = 0$ both row actions vanish. For $k > 0$, put $F = \Delta^{-1/2} E \Delta^{-1/2}$, $h = \Delta^{-1/2} x$, $Q_\perp = I - U_B U_B^\top$. Then $\|F\| \le C/d$ and $\|h\|^2 \le Ck/d$. The scalar-rescaled matrix $Q = G^{-1/2} A G^{-1/2}$ differs from $\mathrm{diag}(0, Q_B)$ by $O(d^{-1/2})$: its off-diagonal row has norm at most $C\sqrt{k/(gd)}$, and the lower renormalization has norm $O(d^{-1})$. Thus the full signal remains separated.

For an eigenvalue μ_r of $Q = \widetilde{A}/\tau_A$, write a full generalized eigenvector as

$$G^{-1/2} u_r = \begin{pmatrix} t_r \\ \Delta^{-1/2}(U_B y_r + z_r) \end{pmatrix}, \qquad U_B^\top z_r = 0.$$

The exact top and complementary equations are

$$h^\top (U_B y_r + z_r) = \mu_r g t_r,$$
$$(\mu_r I - N + \mu_r Q_\perp F Q_\perp) z_r = t_r Q_\perp h - \mu_r Q_\perp F U_B y_r.$$

The inverse T_r of the complementary operator is bounded. Therefore

$$(\mu_r g - (Q_\perp h)^\top T_r Q_\perp h) t_r = (U_B^\top h)^\top y_r - \mu_r (Q_\perp h)^\top T_r Q_\perp F U_B y_r.$$

The denominator has magnitude at least $cd - Ck/d \ge c'd$. Here $g = \max(k, \tau_A) \asymp d$ is the diagonal weight in this formula. The matrix $Y = (y_1, \dots, y_K)$ and its inverse have bounded norm, by the signal-space angle and $(I + F)^{1/2} = I + O(d^{-1})$.

The local row premise and the global threshold perturbation give

$$\left\| U_B^\top h \right\| \le C_R k/\sqrt{nd}, \qquad \|F U_B\|_F \le C_R \sqrt{k}/(d\sqrt{n}) + C/n.$$

The C/n term is retained; it need not be supported on the neighbors. Substitution yields, for $Z_\perp = (z_1, \dots, z_K)$,

$$\|t\| \le C_R/\sqrt{nd}, \qquad \|Z_\perp\| \le C_R\{\sqrt{k}/(d\sqrt{n}) + 1/n\}. \tag{40}$$

Let $V = \Delta^{-1/2}U_B$, $v_r = \Delta^{-1/2}z_r$ and $f_r = Vy_r + v_r$. Exact generalized orthonormality gives

$$Y^\top Y + Z_\perp^\top Z_\perp + gtt^\top + f^\top Ef = I.$$

On the local part of E, use the hit-row bound; on its global part use $\|E_{\text{glob}}\| \le Cd/n$ and $\|f\| \le C/\sqrt{d}$. Consequently

$$\left\|YY^\top - I\right\| \le C_R\{k/(nd) + 1/n\}. \tag{41}$$

For $w_V = V^\top \Delta w_{-i}$, the exact row expansion is

$$(AH_G w)_i = \sum_r (x^\top V y_r + x^\top v_r)\{gt_r w_i + y_r^\top w_V + v_r^\top \Delta w_{-i} + f_r^\top E w_{-i}\}.$$

Only total degree is needed to bound global coefficient vectors. In particular,

$$\begin{aligned}
\left\|x^\top V\right\| &\le C_R k/\sqrt{nd}, & \|w_V\| &\le C_{W_0}\sqrt{nd},\\
\left\|x^\top v\right\| &\le C_R\{k/(d^{3/2}\sqrt{n}) + \sqrt{k}/(n\sqrt{d})\}, & \left\|v^\top \Delta w_{-i}\right\| &\le C_{R,W_0}\{\sqrt{k/d} + \sqrt{d/n}\},\\
\|gt\| &\le C_R\sqrt{d/n}, & \left\|f^\top E w_{-i}\right\| &\le C_{R,W_0}\{k/\sqrt{nd} + \sqrt{d/n}\}.
\end{aligned}$$

For example the global part of the last bound follows from $|f_r^\top E_{\text{glob}} w_{-i}| \le (Cd/n)W_0\sqrt{n}\,\|f_r\| \le C_{W_0}\sqrt{d/n}$. The leading term is $x^\top VYY^\top w_V$; its error relative to $x^\top VV^\top \Delta w_{-i}$ is $O(k^2/(nd) + k/n)$. Every other product is bounded by a fixed multiple of

$$k/d + \sqrt{k/n} + k^2/(nd) + k/n + k\sqrt{k}/(d\sqrt{n}).$$

These quantities are bounded for $k \le Ld$ and $d \le Cn$. This proves the scalar comparison. □

For the empirical cap threshold $\tau = b_0\bar{D}$, the displayed decomposition of E is automatic on the two-sided total-degree event. Indeed, if $\bar{D}_B$ is the mean degree of B, then

$$\tau_A - \tau_B = b_0(2k - \bar{D}_B)/n.$$

First change each deleted degree by x_j while keeping τ_B; this yields $E_{\text{loc}} \in [0, \text{diag}(x)]$. Then change the threshold; this yields $\|E_{\text{glob}}\| \le C_L d/n$.

Corollary D.2 (Exact-degree reconstruction and simple hollowing)**.** *Let R^h be the exact-degree hollow reconstruction in (5). Then $R^h\mathbf{1} = \mathbf{d}$ and $\text{diag}(R^h) = 0$ exactly. For $\bar{w}_{-i} = (n-1)^{-1}\sum_{j\neq i} w_j$, the same local premises give*

$$(R^h w)_i = x^\top\{H_B(w_{-i} - \bar{w}_{-i}\mathbf{1}) + \bar{w}_{-i}\mathbf{1}\} + O_{L,R,W_0}(1). \tag{42}$$

Proof. For $\bar{w} = n^{-1}\sum_j w_j$, exact degree correction gives $(R_1 w)_i = (R_0(w - \bar{w}\mathbf{1}))_i + k\bar{w}$. Apply the preceding lemma. On hit coordinates $|(H_B\mathbf{1})_j| \le C_R$, because $\left\|U_B^\top \Delta^{1/2}\mathbf{1}\right\| \le C\sqrt{nd}$. Replacing $\bar{w}$ by $\bar{w}_{-i}$ therefore costs $O(k/n)$. Moreover (40) gives $\|u_i\|^2 = g\,\|t\|^2 \le C_R/n$, hence $|(R_0)_{ii}| \le C_R d/n$. Also $(R_0\mathbf{1})_i = O_R(k+1)$. Thus $|(R_1)_{ii}| \le C_R d/n$ and the hollow change to a bounded contrast is at most $2W_0|(R_1)_{ii}| = O(d/n)$. The two exact identities follow by summing the defined rows and examining their diagonal. □

The independent coefficient vector in (42) has perturbation $(H_B - I)(w_{-i} - \bar{w}_{-i}\mathbf{1})$. For block-constant contrasts it therefore admits the tilted cancellation in lemma G.1. For clarity, the hit-row exception is paid directly. On the training event, $\Delta_j \ge cd$, so

$$\mathcal{J}_{\text{bad}} := \{j : \left\|\Delta_j^{-1/2}(U_B)_{j\cdot}\right\|_2 > R/\sqrt{nd}\} \subseteq \{j : \sqrt{n}\,\|(U_B)_{j\cdot}\|_2 > R\sqrt{c}\}.$$

Lemma C.1 bounds the expected cardinality of this set on the training event by $Cne^{-c'd\log R}$. Conditioning only on the deleted graph and using $p_{ij} \le Cd/n$ therefore bounds the probability that the tested row hits it by $Cde^{-c'd\log R}$. Choosing a sufficiently large fixed R gives any prescribed fixed exponential rate H. The upper-degree exception $k > Ld$ has arbitrarily large exponential rate after increasing the fixed L. Neither bound uses a union over all possible tested targets.

Uniformity in w is an algebraic stability statement. It does not make full-graph fitted coefficients independent of x. The fitted-score bound first replaces them by deterministic true contrasts; only then is the independent sharp-tail bound applied.

E Block fitting and a uniform refinement basin

Write $D_{0,i} = (\Omega\mathbf{1})_i$ and $\Pi = UU^\top$. For the proof, rescale the encoded adjacency matrix by its common threshold: $Q = \widetilde{A}/\tau = G^{-1/2}AG^{-1/2}$. This changes the eigenvalue scale, not the retained eigenspace. We work on the aggregate event

$$\left\|Q - G^{-1/2}\Omega G^{-1/2}\right\|_{\mathrm{op}} \le C/\sqrt{d}, \qquad \sum_i G_i \le Cnd, \tag{43}$$

$$\sum_i \frac{(G_i-\tau)^2 + (D_i - D_{0,i})^2}{G_i} \le Cn, \qquad \left\|(I-\Pi)G^{1/2}Z\right\|_F \le C\sqrt{n}, \tag{44}$$

$$\max_{a,b} \left|[Z^\top(A-\Omega)Z]_{ab}\right| \le Cn. \tag{45}$$

We also have $G_i \ge \tau \asymp d$, $D_{0,i} \asymp d$, $\bar{D} \asymp d$, and $\tau \ge \max_i D_{0,i}$. The first two lines follow from proposition B.1; the block-total event is paid in the risk proof. All bounds in this section are deterministic on this event. A constant subscript M permits dependence on a fixed basin radius M, but never on the fitted partition or on n.

An aligned population basis is

$$V = G^{-1/2}Z(Z^\top G^{-1}Z)^{-1/2}O,$$

where O is orthogonal. The signal gap gives

$$\|U - V\|_F \le C/\sqrt{d}, \qquad \sqrt{G_i}\,\|V_{i\cdot}\|_2 \le C\sqrt{d/n}. \tag{46}$$

Moreover $\|Q\|_{\mathrm{op}} \le 1$, since $|x^\top Ax| \le \sum_i D_i x_i^2 \le \sum_i G_i x_i^2$. In particular $\|\Xi/\tau\|_{\mathrm{op}} \le 1$, and $R_0 = G^{1/2}Q_K G^{1/2}$ for $Q_K = U(\Xi/\tau)U^\top$.

Lemma E.1 (Weighted mass of incorrectly assigned vertices)**.** *For every set E with $|E| \le Mn/d$,*

$$\sum_{i\in E} G_i \le C_M n, \qquad \sum_{i\in E} \sqrt{G_i}\,\|U_{i\cdot}\|_2 \le C_M\sqrt{n/d}. \tag{47}$$

Also $\sum_i \sqrt{G_i}\,\|U_{i\cdot}\|_2 \le C\sqrt{nd}$. These bounds hold simultaneously for all such sets.

Proof. The inequality $G_i \le 2\tau + (G_i-\tau)^2/G_i$ gives $\sum_E G_i \le C(d|E| + n) \le C_M n$. Put $a_i = \sqrt{G_i}\,\|U_{i\cdot}\|_2$. By (46) and Cauchy–Schwarz,

$$\sum_{i\in E} a_i \le C|E|\sqrt{d/n} + \left(\sum_{i\in E} G_i\right)^{1/2} \|U - V\|_F \le C_M\sqrt{n/d}.$$

The global bound follows from $\sum_i a_i \le (\sum_i G_i)^{1/2}\,\|U\|_F$ and fixed K. □

Lemma E.2 (Radius-count initialization)**.** *The initializer in section 3 makes at most Cn/d errors on the aggregate event, after one common label permutation.*

Proof. The normalized rows of V are the K rows of O, with pairwise distance $\sqrt{2}$. Choose a fixed B_0 larger than twice the uniform upper bound on τ/d. Each row with $G_i > B_0 d$ contributes at least $B_0 d/4$ to $\sum_i (G_i - \tau)^2/G_i$, so there are at most Cn/d such rows. For every remaining row, $\|V_{i\cdot}\|_2 \geq c/\sqrt{n}$. The normalization inequality

$$\left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2 \leq \frac{2\,\|u - v\|_2}{\|v\|_2}$$

(with any fixed unit direction when $u = 0$) and (46) imply that the normalized empirical rows x_i have total squared distance at most Cn/d from their true centers c_{z_i}: the remaining rows satisfy this by the displayed inequality, and the $O(n/d)$ high-degree rows each contribute at most four.

Call a row good if $\|x_i - c_{z_i}\|_2 \leq \epsilon = 1/32$. At most $m = Cn/d$ rows are bad, and $m < \pi_- n/3$ for large n. Inductively, all good rows of a selected block have been removed and all good rows of unselected blocks remain. A radius-1/8 ball centered at a good row of an unselected block contains every good row of that block, hence more than m remaining rows. The maximizing ball must therefore contain a good row from an unselected block; its representative is within $1/8 + \epsilon = 5/32$ of that center. Every good row of that block lies within $3/16 < 1/4$ of the representative. Good rows of any other block are farther than $\sqrt{2} - 3/16 > 1/4$, so deletion preserves the induction. After K selections, each good row is closer to its own representative than to all others. Only the m bad rows can be misclassified. □

Lemma E.3 (Reconstructed block totals)**.** *The hollow, degree-preserving reconstruction in (5) satisfies*

$$\sum_{i,j} |R^h_{ij}| \leq Cnd. \tag{48}$$

For every partition $\widehat{Z}$ with at most Mn/d errors under the common label matching,

$$[\widehat{Z}^\top R^h \widehat{Z}]_{ab} = n_a n_b P_{ab} + O_M(n). \tag{49}$$

The omitted diagonal is included in the remainder.

Proof. Let $Q_{\text{tail}} = (I - \Pi)Q(I - \Pi)$ and $F_1 = (I - \Pi)G^{1/2}\mathbf{1}$. Then $\|Q_{\text{tail}}\|_{\text{op}} \leq C/\sqrt{d}$, $\|F_1\|_2 \leq C\sqrt{n}$, and the degree correction is exactly $v = \mathbf{d} - R_0\mathbf{1} = G^{1/2} Q_{\text{tail}} F_1$. Consequently

$$\sum_i |v_i| \leq Cn, \qquad \sum_i |(R_0)_{ii}| \leq \sum_i G_i \Pi_{ii} \leq 2\tau K + \sum_i \frac{(G_i - \tau)^2}{G_i} \leq Cn.$$

It follows that $\sum_i |(R_1)_{ii}| \leq Cn$, and hollowing has total absolute cost at most $2\sum_i |(R_1)_{ii}|$. Since $|(R_0)_{ij}| \leq a_i a_j$, lemma E.1 yields $\sum_{ij} |(R_0)_{ij}| \leq Cnd$. The correction $v\mathbf{1}^\top/n$ has absolute mass $\sum_i |v_i|$, proving (48).

Put $\widehat{F} = (I - \Pi)G^{1/2}\widehat{Z}$. The wrong-set bound gives $\left\|\widehat{F}\right\|_F \leq C_M\sqrt{n}$. Spectral invariance gives the exact identity

$$\widehat{Z}^\top (R_0 - A)\widehat{Z} = -\widehat{F}^\top Q_{\text{tail}} \widehat{F} = O_M(n/\sqrt{d}).$$

Moving the wrongly labeled vertices changes any raw block total by at most $C\sum_{i \in E} D_i \leq C_M n$. Thus (45) gives (49) for R_0. The degree correction changes a block total by $[\widehat{Z}^\top v]_a \widehat{n}_b/n$, where $\widehat{Z}^\top v = \widehat{F}^\top Q_{\text{tail}} F_1 = O_M(n/\sqrt{d})$. The $O(n)$ hollowing cost completes the proof. □

For a tested vertex i, the local regularity condition is

$$D_i \leq bd, \qquad a_i = \sqrt{G_i}\,\|U_{i\cdot}\|_2 \leq L\sqrt{d/n}, \tag{50}$$

where b, L are fixed. It is imposed only at the tested target.

Lemma E.4 (Local absolute mass)**.** *On* (50)*,*

$$\sum_j |R^h_{ij}| \le C_{b,L} d, \qquad \sup_{|E|\le Mn/d} \sum_{j\in E} |R^h_{ij}| \le C_{M,b,L}. \tag{51}$$

Proof. The bound $|(R_0)_{ij}| \le a_i a_j$ and lemma E.1 give the two claims for R_0. They also imply $|v_i| \le D_i + \sum_j |(R_0)_{ij}| \le Cd$. The rank-one correction therefore adds at most Cd to the full row mass and $Cd|E|/n \le C_M$ to the wrong-set mass. Finally, $|(R_1)_{ii}| \le a_i^2 + |v_i|/n \le Cp$; hollowing changes either bound by at most $2|(R_1)_{ii}|$. □

Proposition E.5 (Block fitting and the Bernoulli contrast)**.** *Uniformly over partitions in the Mn/d basin, the symmetric Bernoulli block fit is unique and interior for large n, and*

$$\max_{a,b} |\widehat{P}_{ab} - P_{ab}| \le C_M/n. \tag{52}$$

At every target satisfying (50)*,*

$$\widehat{L}^{ab}_i = [R^h w_{ab}]_i + b_{i,ab} + O_{M,b,L}(1), \tag{53}$$

where w_{ab} and the full nonedge intercept $b_{i,ab}$ are the true Bernoulli quantities in (13)*.*

Proof. For an unordered fitted block pair, write E_{ab} for its ordered reconstructed total and N_{ab} for the corresponding number of ordered off-diagonal pairs. With $\nu_{ab} = 2 - \mathbf{1}\{a = b\}$, lemma E.3 gives

$$E_{ab} = \nu_{ab} n_a n_b P_{ab} + O_M(n), \qquad N_{ab} = \nu_{ab} \widehat{n}_a \widehat{n}_b + O(n).$$

The exact criterion is $E_{ab} \log q + (N_{ab} - E_{ab}) \log(1-q)$. Since $0 < E_{ab} < N_{ab}$ for large n, it has the unique maximizer $\widehat{P}_{ab} = E_{ab}/N_{ab}$. Its clipping bounds are inactive because this ratio is between fixed positive multiples of p. Using $|\widehat{n}_a - n_a| \le Mn/d$ gives

$$|\widehat{P}_{ab} - P_{ab}| \le C_M\{n/n^2 + p(n/d)/n\} \le C_M/n.$$

The Bernoulli contrast is exactly a log-odds action plus a nonedge intercept. On the positive compact profile class, the fitted log-odds error is $O_M(1/d)$ per coefficient. Its row action costs $O_M(1)$ by (51). Replacing the wrong neighbor labels changes bounded coefficients on at most Mn/d coordinates, again at cost $O_M(1)$. For the intercept, bounded derivatives of $\log(1-q)$ give parameter cost $n\, O_M(1/n) = O_M(1)$. Wrong neighbor labels change each nonedge log-ratio by at most Cp, for an additional cost $C_M(n/d)p = O_M(1)$. All bounds hold simultaneously for every partition in the basin; fitted coefficients need not be independent of the tested row. □

Lemma E.6 (Uniform basin for finite hard refits)**.** *There is a fixed M such that, on the aggregate event and for large n, each block-fit and Bernoulli-assignment update maps the Mn/d error basin into itself under the same label matching. At every target satisfying* (50)*, an error at any updated iterate implies*

$$[R^h w_{z_i b}]_i + b_{i,z_i b} \le C_M \quad \text{for some } b \ne z_i. \tag{54}$$

The same event controls any finite, possibly data-dependent stopping rule after at least one update.

Proof. Choose large fixed b, R, with b larger than twice the uniform upper bounds on $D_{0,i}/d$ and τ/d. The degree energy bounds $|\{i : D_i > bd\}| \le Cn/d$. Also $\|V_{i\cdot}\|_2 \le C/\sqrt{n}$, so (46) gives $|\{i : \sqrt{n}\,\|U_{i\cdot}\|_2 > R\}| \le Cn/d$. Every other target satisfies (50) for fixed L, because $G_i \le C_b d$ there. This exceptional set and its bound are independent of the input partition and of M.

Let $Q_0 = G^{-1/2}\Omega G^{-1/2}$. Best rank-$K$ approximation gives $\|Q_K - Q_0\|_{\mathrm{op}} \le C/\sqrt{d}$; consequently

$$\left\|G^{-1/2}(R_0 - \Omega)Z\right\|_F^2 \le (C/d)\left\|G^{1/2}Z\right\|_F^2 \le Cn.$$

Since $\left\|G^{-1/2}v\right\|_2^2 \le Cn/d$, the degree correction preserves this bound:

$$\left\|G^{-1/2}(R_1-\Omega)Z\right\|_F^2 \le Cn. \tag{55}$$

Hollowing changes the action of any bounded block-constant vector at row i by at most $C|(R_1)_{ii}|$. As $\sum_i |(R_1)_{ii}| \le Cn$, this change exceeds a prescribed small multiple of d on at most Cn/d rows. This auxiliary set is used only to count basin errors, and is not paid as a probability exception in the sharp risk argument.

For $a = z_i$, the population contrast $[\Omega w_{ab}]_i + b_{i,ab}$ is the sum of the incident Bernoulli Kullback–Leibler divergences, with an $O(p)$ completed-diagonal term. Row separation and positive group proportions therefore give a uniform lower bound $c_{\mathrm{gap}}d$. The block coefficients of w_{ab} are bounded. At a regular target, proposition E.5 bounds the fitted discrepancy by C_M, which is at most $c_{\mathrm{gap}}d/4$ for sufficiently large n. Outside the auxiliary hollowing set, any new error thus forces

$$\|[(R_1-\Omega)Z]_{i\cdot}\|_2 \ge cd.$$

For these targets $G_i \le C_b d$. Summing the squared requirement divided by G_i and applying (55) bounds their number by $C_0 n/d$. Including both counting exceptions still gives this form with C_0 independent of M.

Choose M at least C_0 and the initializer constant. The basin is invariant for all sufficiently large n; only that threshold may depend on M. On each regular target, (54) follows directly from (53). It is the same event for every partition in the basin. Induction proves the assertion for finite hard refits and adaptive finite stopping, with no probability union over iterations. □

F Exponential accuracy of the aggregate event

Lemma F.1 (Exponentially accurate aggregate geometry above a logarithmic scale)**.** *Fix any $A_0 > 0$ and $H > 0$. Under the fixed positive compact structural conditions, if $d \ge A_0 \log n$ and $np^2 \le C_1$, there is an aggregate event $\mathcal{E}_H$ with*

$$\mathbb{P}(\mathcal{E}_H^c) \le C_H e^{-Hd}$$

on which the capped-adjacency geometry, weighted degree-energy bounds, two-sided total-degree bounds, and true block-total bounds used in eqs. (43) to (45) hold for the full graph and every graph obtained by deleting at most three vertices. The constants may depend on A_0, H, C_1 and the structural bounds. No condition is imposed on the smallest raw degree or the largest spectral row.

Proof. The centered bounded-entry norm inequality of Bandeira–van Handel [1, Corollary 3.12 and Remark 3.13] gives

$$\mathbb{P}\{\|A-\Omega\|_{\mathrm{op}} > L_H\sqrt{d}\} \le e^{-(H+2)d}$$

after choosing a sufficiently large fixed L_H. Indeed, the row variance sum is $O(d)$, the entry bound is one, and the prefactor n in the norm-tail inequality is absorbed using $\log n \le d/A_0$. The completed-mean diagonal has norm $O(p)$ and is absorbed in the same threshold. No minimum raw degree is imposed.

The Bernoulli transform of the total number of edges gives, with failure probability at most Ce^{-cnd},

$$cnd \le \sum_i D_i \le Cnd.$$

For each of the fixed K^2 true block totals, Bernstein's inequality for the independent upper-triangular edges, whose coefficients have absolute value at most two, gives

$$\mathbb{P}\left\{\left|[Z^\top(A-\Omega)Z]_{ab}\right| > Ln\right\} \le 2\exp\left\{-\frac{cL^2n}{d+L}\right\}.$$

The deterministic completed diagonal costs $O(d)$ and is included by increasing L. Since $d \to \infty$ and $d^2/n \le C_1$, the exponent is at least $c'L^2d/C_1$ for all sufficiently large n. Increasing the fixed $L = L_H$ therefore makes the failure of all block totals at most $C_H e^{-(H+2)d}$. The total-degree failure is also bounded by $Ce^{-(H+2)d}$ for large n.

These three full-graph events imply the required events on every fixed-depth minor without a degree-wise probability union. To see this, let $J = [n] \setminus T$ with $|T| \le 3$, set $A_J = A_{JJ}$ and $\Omega_J = \Omega_{JJ}$, and recompute all degrees and the cap threshold on J. The centered matrix $A_J - \Omega_J$ is a principal submatrix of $A - \Omega$, so its norm is at most $L_H\sqrt{d}$. Removing T changes the total raw degree by at most $2|T|n$; as $d \to \infty$, the two-sided bound by fixed multiples of nd therefore persists uniformly over T. Removing T changes a true block total of $A - \Omega$ by $O(|T|n)$ deterministically, since the entries have bounded absolute value. Its $O_H(n)$ bound persists as well. The surviving group masses differ from their original values by $O(1)$ and remain comparable to n.

The recomputed average degree is consequently comparable to d. For the fixed sufficiently large cap multiplier b_0, its threshold $\tau_J = b_0\overline{D}_J$ lies between positive fixed multiples of d and above every coordinate of $D_{0,J} = \Omega_J\mathbf{1}$. Thus $G_{J,j} = \max(D_{J,j}, \tau_J) \ge cd$ and $\sum_j G_{J,j} \le Cnd$. The raw norm bound immediately gives

$$\left\|G_J^{-1/2}(A_J - \Omega_J)G_J^{-1/2}\right\|_{\mathrm{op}} \le C_H d^{-1/2},$$
$$\sum_{j\in J} \frac{(D_{J,j} - D_{0,J,j})^2}{G_{J,j}} = \left\|G_J^{-1/2}(A_J - \Omega_J)\mathbf{1}\right\|_2^2 \le C_H n.$$

The cap exceeds the completed population degrees, so coordinatewise

$$\frac{(G_{J,j} - \tau_J)^2}{G_{J,j}} \le \frac{(D_{J,j} - D_{0,J,j})^2}{G_{J,j}}.$$

Indeed, the left side is zero when $D_{J,j} \le \tau_J$; otherwise $0 \le D_{J,j} - \tau_J \le D_{J,j} - D_{0,J,j}$. This proves the other weighted energy bound. The order-d signal gap of the capped adjacency matrix (a fixed gap after dividing by τ_J) and the residual $\left\|(I - U_J U_J^\top)G_J^{1/2}Z_J\right\|_F \le C_H\sqrt{n}$ now follow by the deterministic Gram-matrix and projection arguments in proposition B.1. Those arguments use only $G_{J,j} \ge \tau_J \asymp d$, total degree, the displayed rescaled noise bound, and the energy relative to τ_J, all just proved. Intersecting the stated aggregate consequences defines $\mathcal{E}_H$. The three full-graph events imply it simultaneously for every indicated minor, proving its claimed failure bound.

For a deleted graph used as independent training data, form the analogous event using that graph and its further minors. The same argument gives failure probability $C_H e^{-Hd}$, and this event is measurable from the training graph alone. The full-graph common event is used only for deterministic implications, never as a condition that would preserve independence of an incident row. □

G Completion of the oracle-risk bound

The final step uses a deleted graph only as a proof device. The implemented procedure retains one full-graph eigensystem. For each target i, put $B = A_{-i,-i}$ and recompute its degrees, cap threshold, weights and projection H_B. Throughout this section, all constants are uniform over the fixed structural class.

Lemma G.1 (Ordinary Bernoulli tilted cancellation)**.** *Let w be a true block-constant Bernoulli log-odds vector on the deleted graph and put*

$$f = w - \bar{w}\mathbf{1}, \qquad e = (H_B - I)f, \qquad \bar{w} = (n-1)^{-1}\mathbf{1}^\top w.$$

On the deleted aggregate event, under the exact Chernoff tilt of the raw-row oracle contrast, its Bernoulli probabilities r_j satisfy

$$\left|\sum_{j\neq i} r_j e_j\right| \le C, \qquad \sum_{j\neq i} r_j e_j^2 \le C. \tag{56}$$

Proof. The tilted probabilities r_j are constant within each true block and satisfy $r_j \le Cp$. Write τ and G for the threshold and diagonal matrix recomputed from B. Then $r = \tau h$ exactly for the block-constant vector $h = r/\tau$, with $\|h\|_\infty \le C/n$ because $\tau \asymp d = np$. The common scalar τ may depend on the training graph; this identity holds pointwise on its aggregate event. Since H_B is the G-orthogonal projector,

$$r^\top e = -\langle (I - H_B)h, (I - H_B)f\rangle_G + h^\top(\tau I - G)e.$$

By (20), the two norms in the first term are at most $C/\sqrt{n}$ and $C\sqrt{n}$. For the second term, $\sum_j (G_j - \tau)^2/G_j \le Cn$ gives

$$\left\|G^{-1/2}(\tau I - G)h\right\|_2 \le C/\sqrt{n}, \qquad \|e\|_G \le C\sqrt{n}.$$

This proves the first bound. Since $r_j \le Cp$ and $\|e\|_2^2 \le Cn/d$, the second bound follows as well. The deterministic nonedge intercept remains in the oracle score and is not absorbed into this cancellation. When the cap is inactive on the deleted graph, $G = \tau I$ and $H_B = \Pi_B = U_B U_B^\top$. The second term vanishes, leaving the ordinary Euclidean adjacency identity

$$r^\top e = -\langle (I - \Pi_B)r, (I - \Pi_B)f\rangle.$$

□

Proof of theorem 4.1. Fix $A_0 > 0$. First consider $d \le A_0 \log n$, with $d \to \infty$. Let $\mathcal{E}$ collect the aggregate conditions (43)–(45), two-sided total degree, and their recomputed versions after deletion of at most three vertices. For each prescribed fixed $r > 0$, proposition B.1 gives these geometric conditions with failure probability at most $C_r n^{-r}$, by increasing its probability parameter to cover the polynomial number of minors. For any true block total, independent-edge Bernstein gives

$$\mathbb{P}\{|[Z^\top(A - \Omega)Z]_{ab}| > Ln\} \le 2\exp\{-cL^2 n/(d + L)\}.$$

In the present range this is smaller than any fixed inverse power of n, as is the total-degree failure. The same holds for the required minors. For a target i, define $\mathcal{E}_i$ using only its deleted graph and its further deletions. Then

$$\varepsilon_{\text{geom}} := \mathbb{P}(\mathcal{E}^c) + \frac{1}{n}\sum_i \mathbb{P}(\mathcal{E}_i^c) \le C_r n^{-r}. \tag{57}$$

On $\mathcal{E}$, lemmas E.2 and E.6 place every updated iterate in one fixed Mn/d basin. At target i impose only $D_i \le bd$, its local spectral-row condition (50), and the requirement that the incident row hits no exceptional deleted spectral row. By lemma C.1 and corollaries C.2 and C.3, the average probability cost of these spectral exceptions is at most $C_H e^{-Hd}$ for any prescribed fixed H, on choosing fixed thresholds sufficiently large. Indeed, $G_j \ge \tau \asymp d$ gives (38) outside a deleted exceptional set $\{j : \sqrt{n}\,\|(U_B)_{j\cdot}\|_2 > R\}$, whose expected hit mass is controlled by (36). The degree tail has the same bound by the single-row Bernoulli Chernoff inequality. Polynomial factors in d are absorbed by first choosing a larger exponential rate. These are averages of target probabilities, without a union over all tested vertices.

On the resulting local events, the deterministic fitted-score bound and corollary D.2 imply that an aligned error at i requires, for some $b \neq a = z_i$,

$$T_{i,ab} + \sum_{j \neq i} A_{ij} e_j \leq C_M, \qquad e = (H_B - I)(w_{ab,-i} - \bar{w}_{ab,-i}\mathbf{1}). \tag{58}$$

Here $T_{i,ab}$ is the exact raw Bernoulli oracle contrast, including $b_{i,ab}$. The coefficients e are measurable from B and independent of $(A_{ij})_{j \neq i}$. To apply the probability bound, first use this deterministic inclusion, then drop the full-graph event and condition only on B on $\mathcal{E}_i$. No independence is asserted conditional on $\mathcal{E}$.

The tilted mean and quadratic bounds in lemma G.1 give the moment hypotheses of lemma A.2. Its exceptional-coefficient hit term is at most $C_H e^{-Hd}$ by corollary C.2, after selecting a fixed coefficient threshold. That truncation is used only in the proof. Applying the sharp transfer to (58), averaging targets and taking the fixed-K union over competitors, we obtain

$$\mathbb{E}\ell(\widehat{z}, z) \leq \frac{C}{n} \sum_i \sum_{b \neq z_i} d^{-1/2} e^{-I_{i,z_i b}} + C_H e^{-Hd} + \varepsilon_{\text{geom}}. \tag{59}$$

The strict and non-strict oracle comparisons in (15) bound the first term by $CR_{\text{SBM},*}(n)$, independently of the oracle tie rule.

On the compact structural class $I_{i,ab} \leq C_I d$. Hence $R_{\text{SBM},*}(n) \geq c d^{-1/2} e^{-C_I d}$. Choose $H > C_I + 1$ and $r > A_0(C_I + 1) + 2$. When $d \leq A_0 \log n$,

$$\frac{e^{-Hd} + n^{-r}}{d^{-1/2} e^{-C_I d}} \longrightarrow 0,$$

which pays both remainders in (59).

For $d > A_0 \log n$, use lemma F.1 in place of the polynomially accurate geometry. It gives $\varepsilon_{\text{geom}} \leq C_{H'} e^{-H'd}$ for any fixed $H' > 0$, with training events still measurable from the respective deleted graphs. The row-tail proof uses deterministic aggregate geometry and applies innovation concentration before restricting to that event. Its estimates, the local LOO comparison and the fitting argument therefore remain valid for this stronger event. Equation (59) follows with the new geometry remainder. Choosing $H, H' > C_I + 1$ pays both exponential terms. The two proof ranges cover every sequence $d \to \infty$ under $np^2 \leq C_1$, proving the theorem without a logarithmic degree lower bound. □

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